Reverse Linked List

AlgoExpert · Hard · Linked Lists

The problem

- Reverse in place and return the new head.

Intuition

- Flip each link to point backwards, keeping three pointers so nothing is lost.

Approach

- 1. Previous starts null
- 2. Save next, point current at previous
- 3. Advance both
- 4. Previous is the new head

Java solution

```java
public class ReverseLinkedList {
    public static LinkedListNode reverseLinkedList(LinkedListNode head) {
        LinkedListNode prev = null;
        while (head != null) { LinkedListNode n = head.next; head.next = prev; prev = head; head = n; }
        return prev;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Flip each link to point backwards, keeping three pointers so nothing is lost.
- Complexity: O(n)